Handout

Physics Grade 9 - Dynamic Electricity

Dynamic electricity is a form of electricity that occurs due to the movement of electric charges, usually electrons, through a conductor such as a metal wire. This movement of charges is known as electric current. Dynamic electricity is commonly found in everyday electrical devices like lamps, fans, and televisions, where electric energy flows continuously to make the devices work.

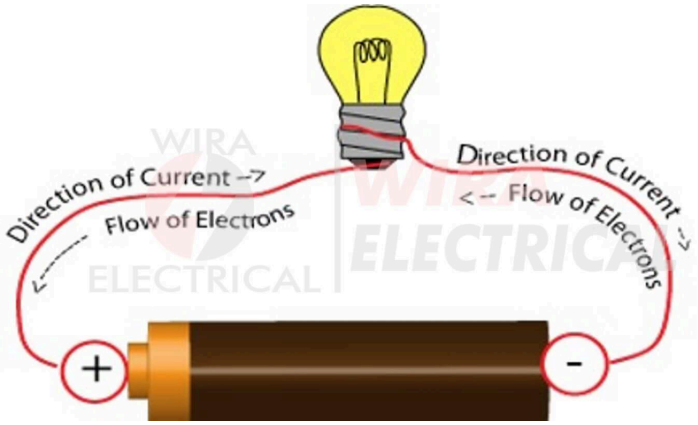

Fig 1. Electric current flows from the positive to the negative terminal, while electrons move in the opposite direction, lighting the bulb as they pass through the circuit.

Conductor and Insulator

A conductor is a material that allows electric current to flow easily because it has many free electrons. Metals such as copper, aluminum, and iron are good conductors and are commonly used in electrical wires and circuits.

An insulator is a material that does not allow electric current to flow easily because its electrons are tightly bound. Materials such as rubber, plastic, glass, and wood are insulators and are used to cover wires or protect people from electric shock.

Fig 3. Example of conductors and insulators

Electric Current

Electric current is defined as the amount of electric charge flowing through a conductor per unit time. The greater the number of charges that move in a certain amount of time, the larger the current. This relationship is presented in the formula:

$$I = \frac{Q}{t}$$

The symbol Q represents electric charge, which is the amount of charge that moves through the circuit and is measured in coulombs (C). The symbol t represents time, which is the duration during which the charge flows and is measured in seconds (s).

Electric current, symbolized by I, is measured in **amperes (A)**. In electrical circuits, current flows from the positive terminal to the negative terminal of a power source through the circuit. An ammeter could be used to measure the current by connecting it in series with the component or branch of the circuit that you want to measure.

Electric Potential Difference / Voltage

Voltage, also known as electric potential, is the electrical energy possessed by a charge at a certain point in an electric field. It shows how much energy is available to move electric charges. Voltage is defined as the electric potential energy per unit charge and can be written using the formula:

$$PE = V \times Q$$

where PE is electric potential energy, V is voltage, and Q is charge. Voltage is measured in **volts (V)**, where one volt means one joule of energy per coulomb of charge.

In an electric field, a positive charge naturally moves from a higher potential to a lower potential without doing work, while moving a charge from lower potential to higher potential requires work. In electrical circuits, voltage is measured using a voltmeter, which is connected in parallel to the component being measured.

Electrical Resistance

Electrical resistance is the property of a material that resists the flow of electric current. Materials with low resistance allow electric current to flow easily, while materials with high resistance limit the flow of current. Resistance is measured in **ohms (Ω).**

Ohm's Law

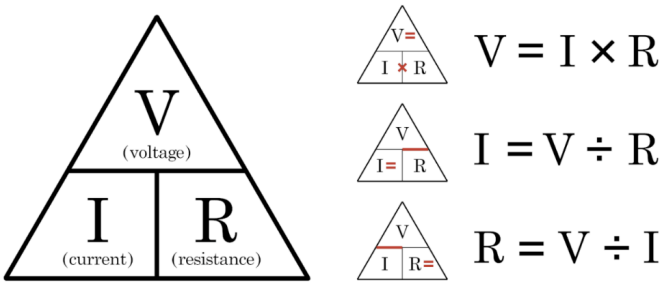

Fig 3. Ohm's Law formula

The relationship between voltage, current, and resistance is described by Ohm's Law, which states that the current flowing through a conductor is directly proportional to the voltage across it and inversely proportional to its resistance. This relationship helps explain how changes in voltage or resistance affect the current in an electrical circuit.

Circuits

Series Circuit

In a series circuit, components are connected in a single path, so the same current flows through all components. However, the voltage is shared between the components, so each component gets only part of the total voltage. The total resistance in a series circuit is found by adding all the resistances

together, which means the total resistance becomes larger when more components are added. If one component fails, the entire circuit stops working.

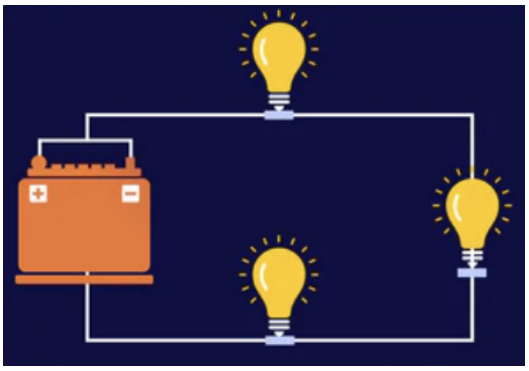

Fig 4. Series circuit

Parallel Circuit

In a parallel circuit, components are connected in separate branches, allowing current to flow through multiple paths. In this type of circuit, the voltage is the same across all components, equal to the voltage of the power source. Because electricity can choose multiple routes instead of just one, it can flow more easily overall. This is why the total resistance becomes lower, as adding more paths reduces the difficulty for current to move through the circuit. In a parallel circuit, if one component fails, the others can continue to operate.

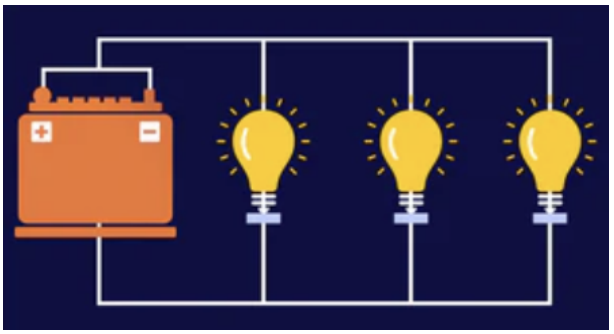

Fig 5. Parallel circuit